A Direct FEM-BIE Coupling for Free-Space Helmholtz Scattering

Graham Mauer[1] Peter Munch[2] Adrianna Gillman[1]

[1] University of Colorado Boulder [2] Technical University of Berlin

Overview

Free-space scattering problems involving locally compact, non-homogeneous media arise in a variety of applications including medical imaging, seismology, and acoustics. These problems are challenging because it is impossible to discretize all of space. Conventional solution methods, such as absorbing boundary conditions and perfectly matched layers, truncate the domain but sometimes give rise to artificial reflections especially as the frequency grows. We propose a finite element (FEM)-based technique that utilizes integral equations to naturally satisfy the radiation condition and mitigate such reflections. An additional advantage of this technique is that the discretizations of both the integral equation and the FEM only need to be constructed once, independent of the number of incident waves.

Physical Problem

We consider the following free-space Helmholtz scattering problem for time-harmonic waves in two spatial dimensions.

$$\begin{cases} \Delta u^{\mathrm{s}} + \kappa^2(1 - b(\boldsymbol{x}))u^{\mathrm{s}} = \kappa^2 b(\boldsymbol{x})u^{\mathrm{i}}, \quad \boldsymbol{x} \in \mathbb{R}^2, \\ \dfrac{\partial u^{\mathrm{s}}}{\partial r} - i\kappa u^{\mathrm{s}} = o(r^{-1/2}), \quad r := |\boldsymbol{x}| \to \infty. \end{cases} \tag{1}$$

Here, u^{s} and u^{i} are the scattered and incident components of the total field, u, such that $u = u^{\mathrm{s}} + u^{\mathrm{i}}$, $b(\boldsymbol{x})$ is a smooth compactly supported function, $\kappa \in \mathbb{R}^+$ is the wavenumber, and the second condition is the Sommerfeld radiation condition.

Artificial Boundary, FEM Mesh, and BIE Nodes

We split $\mathbb{R}^2$ into interior and exterior regions Ω and Ω^c as shown in Figure 1. Ω is discretized using finite elements, so we consider the weak form of (1), with Dirichlet data on the boundary in the place of the radiation condition. The exterior region is treated with boundary integral equations (BIEs) with the operators defined on the Gaussin nodes in figure (1).

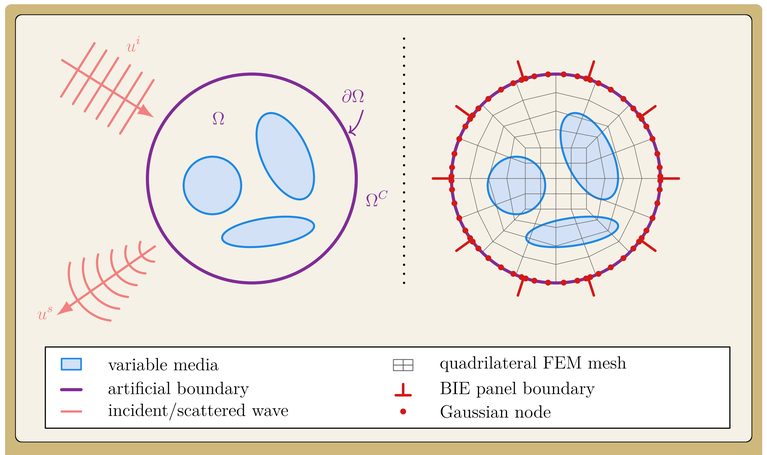


Figure 1: Artificial boundary, FEM mesh, and BIE nodes.

Our procedure heavily utilizes general-purpose algorithms from the finite-element library `deal.II` for processes related to the interior domain [1]. As a result, the respective code is compact and relatively readable. This will allow our source code to be easily adopted by current users of the `deal.II` library.

Coupling Procedure

The coupling enforces continuity of the FEM and BIE normal fluxes across the artificial boundary through Dirichlet-to-Neumann (DtN) operators. The interior and exterior DtN operators, T_{int} and T_{ext}, map interior/exterior Dirichlet data to interior/exterior Neumann data. The coupling is formatted in a well conditioned [2] manner as follows:

$$\left(\frac{1}{2}I - D + ST_{\mathrm{int}}\right) u^{\mathrm{s}}|_{\partial\Omega} = S\left(u^{\mathrm{i}}_n - T_{\mathrm{int}}u^{\mathrm{i}}|_{\partial\Omega}\right). \tag{2}$$

Coupled FEM–BIE Algorithm

1 **Construct the interior DtN operator,** T_{int}**,** from FEM discretization on Ω.
Assemble the FEM operator using the weak form of (1) such that

$$\mathcal{A}_{i,j} = \int_\Omega -\nabla\varphi_i \cdot \nabla\varphi_j + \kappa^2(1 - b(\boldsymbol{x}))\varphi_i\varphi_j \, dV, \qquad b_j = \int_\Omega \kappa^2 b(\boldsymbol{x})u^{\mathrm{i}}\varphi_j \, dV. \tag{3}$$

Permute interior and boundary DOFs to obtain the block system

$$\begin{bmatrix} \mathcal{A}_{II} & \mathcal{A}_{IB} \\ \mathcal{A}_{BI} & \mathcal{A}_{BB} \end{bmatrix} \begin{bmatrix} u^{\mathrm{s}}_I \\ u^{\mathrm{s}}|_{\partial\Omega} \end{bmatrix} = \begin{bmatrix} b_I \\ b_B \end{bmatrix} \implies \mathcal{A}_{II}[u^{\mathrm{s}}_I] = b_I - \mathcal{A}_{IB}[u^{\mathrm{s}}|_{\partial\Omega}] \tag{4}$$

Form the sparse block normal derivative operators, $\mathcal{N}_{BB}$ and $\mathcal{N}_{BI}$.

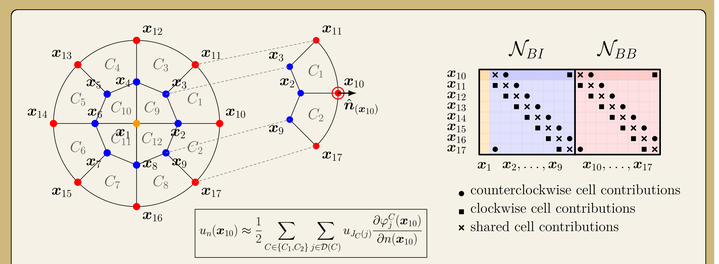


Figure 2: **Left:** Example of a toy mesh with degree 1 shape functions and stencil for computing the normal derivative at $\boldsymbol{x}_{10}$. **Right:** Example of sparse normal operators resulting from the toy mesh.

Combine the homogeneous solution to (4) with $\mathcal{N}_{BI}$ and $\mathcal{N}_{BB}$, to compute the interior DtN map:

$$T_{\mathrm{int}} = \mathcal{N}_{BB} - \mathcal{N}_{BI}\mathcal{A}_{II}^{-1}\mathcal{A}_{IB}. \tag{5}$$

2 **Build BIE operators** D, $\mathscr{D}$, S, and $\mathscr{S}$.
Any radiative solution u^{s} of the Helmholtz equation in Ω^c satisfies [3]

$$u^{\mathrm{s}}(\boldsymbol{x}) = \left(\mathscr{D}\, u^{\mathrm{s}}|_{\partial\Omega}\right)(\boldsymbol{x}) - \left(\mathscr{S}\, u^{\mathrm{s}}_n|_{\partial\Omega}\right)(\boldsymbol{x}), \qquad \boldsymbol{x} \in \Omega^c, \tag{6}$$

where

$$(\mathscr{D}\phi)(\boldsymbol{x}) := \int_{\partial\Omega} \frac{\partial}{\partial n_{\boldsymbol{y}}} \left(\frac{i}{4}H_0^{(1)}(\kappa|\boldsymbol{x} - \boldsymbol{y}|)\right) \phi(\boldsymbol{y}) \, ds_{\boldsymbol{y}}, \quad (\mathscr{S}\phi)(\boldsymbol{x}) := \int_{\partial\Omega} \frac{i}{4}H_0^{(1)}(\kappa|\boldsymbol{x} - \boldsymbol{y}|) \, \phi(\boldsymbol{y}) \, ds_{\boldsymbol{y}}. \tag{7}$$

$H_0^{(1)}$ is the outgoing zeroth-order Hankel function of the first kind and $n_{\boldsymbol{y}}$ denotes the outward unit normal at $\boldsymbol{y} \in \partial\Omega$. Operators D and S result from the limit of $\mathscr{D}$ and $\mathscr{S}$ as $\boldsymbol{x} \to \partial\Omega$ [3].

3 **Determine the unique Dirichlet data on the boundary,** $u^{\mathrm{s}}|_{\partial\Omega}$, using (2).

4 **Solve interior problem** using (4) and **reconstruct exterior solution** using (6).

Effects of Circle Geometry

- **Curved boundary representation.** The artificial boundary must remain geometrically accurate under mesh refinement.
  Solution: use a high-order polynomial representation of the circular boundary so that the FEM geometry matches the boundary used by the BIE.
- **Accurate normal derivatives.** Small errors in the boundary geometry or unit normal directly affect the FEM normal derivative and therefore the interior DtN map.
  Solution: compute normals using the curved mapping and construct the normal derivative operator directly from FEM basis-function gradients (Figure 2). We average repeated contributions for stability.
- **Different boundary discretizations.** FEM boundary degrees of freedom and BIE Gauss nodes do not generally coincide.
  Solution: construct interpolation operators between the two discretizations so the different nodes can communicate.

Sparse Linear Algebra

- **Exploiting sparsity and avoiding a dense inverse.** The interior DtN operator contains $\mathcal{A}_{II}^{-1}$, while $\mathcal{A}_{II}$ may contain millions of interior degrees of freedom. Constructing T_{int} requires solving the matrix equation
  $$\mathcal{A}_{II}X = \mathcal{A}_{IB}$$
  and thus requires solves for many right-hand side vectors.
  Solution: Compute and reuse the sparse LU factorization of $\mathcal{A}_{II}$ and process the right-hand sides in batches. We also reuse the factorization in step 4 above.
- **Memory usage.** Storing all intermediate solution vectors simultaneously can become expensive for large FEM systems.
  Solution: exploit the sparsity of $\mathcal{A}_{II}$, $\mathcal{A}_{IB}$, and $\mathcal{N}_{BI}$ and only assemble the small boundary-to-boundary DtN operator densely.

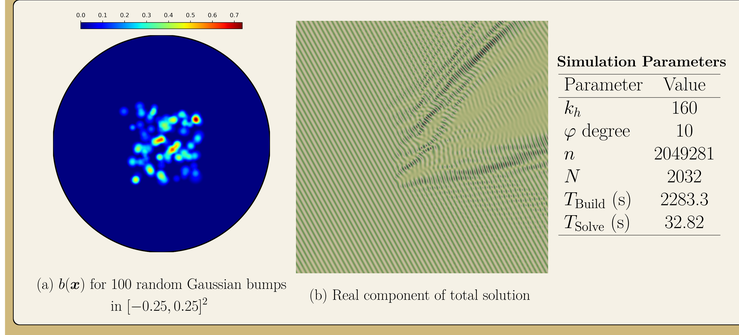
Numerical Results

Symmetric Gaussian: $\kappa = 40$, $b(\boldsymbol{x}) = -1.5\exp(-160|\boldsymbol{x}|^2)$

Deg. φ	n	N	(x, y)	rel. err.	Build (s)	Solve (s)
2	82177	1008	$(0, 0.5)$	1.4×10^{-2}	17.07	1.082
2	82177	1008	$(1, 0)$	4.7×10^{-3}		
2	82177	1008	$(1, 0.5)$	6.2×10^{-3}		
4	328193	1008	$(0, 0.5)$	1.5×10^{-5}	96.89	2.104
4	328193	1008	$(1, 0)$	4.9×10^{-6}		
4	328193	1008	$(1, 0.5)$	5.5×10^{-6}		
6	738049	1008	$(0, 0.5)$	9.2×10^{-9}	342.7	3.946
6	738049	1008	$(1, 0)$	1.8×10^{-9}		
6	738049	1008	$(1, 0.5)$	3.4×10^{-9}		
8	1311745	1008	$(0, 0.5)$	1.9×10^{-11}	959.2	6.382
8	1311745	1008	$(1, 0)$	1.3×10^{-10}		
8	1311745	1008	$(1, 0.5)$	6.1×10^{-11}		

Table 1: Relative errors and build/solve times for the symmetric Gaussian problem with $\kappa = 40$ as the finite-element polynomial degree is increased. Hereafter, n is the number of total FEM DOFs and N is the number of BIE nodes. $(0, 0.5)$, $(1, 0)$, and $(1, 0.5)$ lie in the exterior, on the boundary, and in the exterior, respectively.

Challenging Scattering Example: 100 Random Gaussian Bumps



(a) $b(\boldsymbol{x})$ for 100 random Gaussian bumps in $[-0.25, 0.25]^2$

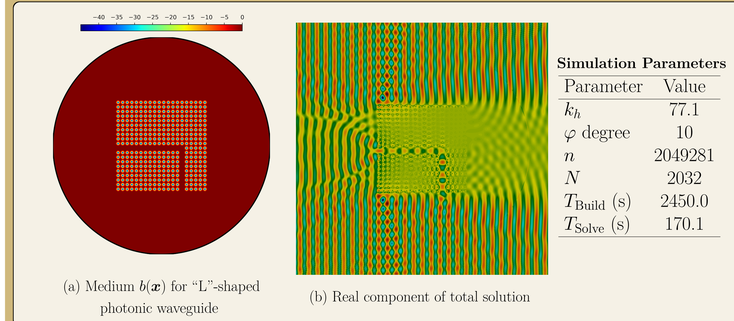
(b) Real component of total solution

Simulation Parameters

Parameter	Value
k_h	160
φ degree	10
n	2049281
N	2032
T_{Build} (s)	2283.3
T_{Solve} (s)	32.82

Figure 3: Scattering off a collection of 100 random Gaussian bumps.

Challenging Scattering Example: Near Resonant Waveguide



(a) Medium $b(\boldsymbol{x})$ for "L"-shaped photonic waveguide

(b) Real component of total solution

Simulation Parameters

Parameter	Value
k_h	77.1
φ degree	10
n	2049281
N	2032
T_{Build} (s)	2450.0
T_{Solve} (s)	170.1

Figure 4: Scattering off an "L"-shaped waveguide at a near-resonant frequency. Rel. error at $(1, 0)$ is 1.0×10^{-7} by comparison to [2].

References

[1] D. Arndt et al., "The deal.II library, version 9.7," *Journal of Numerical Mathematics*, vol. 33, no. 4, pp. 403–415, 2025. DOI: 10.1515/jnma-2025-0115.

[2] A. Gillman, A. H. Barnett, and P.-G. Martinsson, "A spectrally accurate direct solution technique for frequency-domain scattering problems with variable media," *BIT Numerical Mathematics*, vol. 55, pp. 141–170, 2015.

[3] R. Kress, "Linear integral equations,", pp. 58–71, 1999.

Contact: `graham.mauer@colorado.edu`